Fiscal Year:	FY 2020	**Task Last Updated:**	FY 06/08/2019
PI Name:	Yagoobi, Jamal Ph.D.		
Project Title:	Electrically Driven Liquid Film Boiling: A Two-Phase Heat Transport Device Driven by Electric Conduction Mechanism		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Worcester	**State:**	MA
Zip Code:	01609-2280	**Congressional District:**	3
Comments:	NOTE PI formerly at Illinois Institute of Technology; moved to Worcester Polytechnic Institute in January 2012. Info per PI (Ed., 1/17/2012)		
Project Type:	Flight	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	10/01/2016	**End Date:**	09/30/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Hasan, Mohammad	**Contact Phone:**	216-977-7494
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:	June 2019 report: No change to report.		
COI Name (Institution):	Didion, Jeffrey M.S. (NASA Goddard Space Flight Center)		
Grant/Contract No.:	NNX16AT09G		
Performance Goal No.:			
Performance Goal Text:			
	NOTE: Continuation of "Electrically Driven Liquid Film Boiling: A Two-Phase Transport Device Driven By Electric Conduction Mechanism," grant NNX12AR32G, with same Principal Investigator (PI) Jamal Yagoobi, Ph.D. Electrically Driven Liquid Film Boiling - A Two-Phase Heat Transport Device Driven by Electric Conduction Mechanism is an International Space Station (ISS) Experiment to investigate the effects of gravity on: 1.) electrically generated two-phase flow and 2.) electrically driven liquid film boiling. The proposed work is a collaborative effort between the investigators to develop and validate the flight experiment hardware. We propose this effort based upon the successes of our current project: NASA Grant/Cooperative Agreement: NNX12AR32G Electrically Driven Liquid Film Boiling - A Two-Phase Heat Transport Device Driven by Electric Conduction Mechanism. Although the original effort extends through February 2017, we have completed tasks and achieved the technical and programmatic milestones of		

Task Description:	this project. That is, we have passed all the requirements of the Science Concept Review and received the authorization to advance to the next stage of the experiment development. Therefore, the PI and Co-PI submit this proposal to develop the flight experiment hardware with an effective start date of October 2016. This is a proposal to support the development of the experiment flight hardware for the International Space Station, conduct the experiment on–board the International Space Station, support the operations of the experiment and evaluate the experimental data. The proposed period of performance is October 2016 to September 2021. The robust collaboration between Worcester Polytechnic Institute (WPI) and Goddard Space Flight Center (GSFC) can provide both state of the art theoretical investigation and deliver flight qualified experimental hardware; thus, this proposal presents both investigative and hardware development tasks. The PI will maintain overall technical direction of the experiment development; however, the Co-I will support the flight experimental hardware development in collaboration with Glenn Research Center. Note: This project successfully passed all requirements of the Science Concept Review and has received authorization to advance to the next stage of development. This is to elevate it to the International Space Station flight status as of October 1, 2016, with this continuation grant.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	There are a number of potential terrestrial and space applications for this electrohydrodynamic (EHD) driven two-phase heat transport device in all scales (from macro-scale to truly micro-scales). For example, this device can be used to effectively remove heat from electronics in micro- and zero-gravity environment for NASA's future space stations and long term missions. It can also be used in satellites (e.g., embedded system into satellite's multi-functional structure). Terrestrially, it can be used in high intensity electronics cooling as well as in various phase change devices.
Task Progress:	The WPI team has been continuously assisting the NASA GSFC team in developing the hardware for the ISS project scheduled for 2021. The WPI team has also been assisting the NASA GSFC in developing the flight hardware for the upcoming zero-gravity flight scheduled for the week of August 5, 2019. The WPI team has made significant progress in developing flexible EHD conduction pumping technology. This makes the EHD pumping technology very attractive for various terrestrial and reduced gravity applications. The design and performance evaluation of flexible Electrohydrodynamic (EHD) pump, based on EHD conduction pumping technology in macro- and meso-scales, was completed. Unlike mechanical pumps, EHD conduction pumps have no moving parts, can be easily adjusted to the micro-scale, and have been shown to generate and control the flow of refrigerants for electronics cooling applications. However, these pumping devices have only been previously tested in rigid configurations. In 2018 and 2019, for the first time, the net flow generated by flexible EHD conduction pumps was measured on a flat-plane and in various bending configurations. In this behavioral characteristics study, the results show that the flexible EHD conduction pumps are capable of generating significant flow velocities in all size scales (from macro to micro) considered in this study, with and without bending. This study also proves the viability of screen printing as a manufacturing method for these pumps. Finally, a full set of experiments were carried out with series of hydrofluoroether (HFE) dielectric fluids to determine their corresponding electrically driven liquid film flow boiling characteristics, in the presence and absence of the dielectrophoretic force mechanism. The tests were conducted using the test cell developed for the upcoming ISS experiments.
Bibliography Type:	Description: (Last Updated: 01/30/2026)
Abstracts for Journals and Proceedings	Castaneda AJ, O'Connor NJ, Yagoobi J. "Experimental Study of Flexible EHD Conduction Pumping for Electronics Cooling." Presented at the 2019 Annual Meeting of the Electrostatics Society of America, Rochester, NY, June 10-12, 2019. 2019 Annual Meeting of the Electrostatics Society of America, Rochester, NY, June 10-12, 2019. , Jun-2019
Articles in Other Journals or Periodicals	Vazquez PA, Talmor M, Seyed-Yagoobi J, Traore P, Yazdani M. "In-depth description of electrohydrodynamic conduction pumping of dielectric liquids: Physical model and regime analysis." Physics of Fluids. In review as of August 2019. , Aug-2019
Awards	Yagoobi J. (Jamal Yagoobi) "IEEE International Conference on Dielectric Liquids Hans Tropper Memorial Lecture and Award (in Rome, Italy), July 2019." Jul-2019
Papers from Meeting Proceedings	Tobar NV, Christidis PN, O'Connor NJ, Talmor M, Yagoobi J. "Experimental Study of Flexible Electrohydrodynamic Conduction Pumping for Electronic Cooling." ASME 2018 International Technical Conference and Exhibition on Packaging and Integration of Electronic and Photonic Microsystems, San Francisco, California, August 27–30, 2018. ASME Proceedings, ASME 2018 International Technical Conference and Exhibition on Packaging and Integration of Electronic and Photonic Microsystems. Paper No. IPACK2018-8322. https://doi.org/10.1115/IPACK2018-8322 , Aug-2018
Papers from Meeting Proceedings	Duran-Olivencia FJ, Yagoobi J. "The Effect of Substrate Curvature on Flexible EHD Conduction Pumping Performance – A Numerical Study." IEEE 2nd International Conference on Dielectrics, Budapest, Hungary, July 1-5, 2018. 2018 IEEE 2nd International Conference on Dielectrics (ICD). p. 802-807. https://doi.org/10.1109/ICD.2018.8468475 , Jul-2018
Papers from Meeting Proceedings	Yagoobi JS. "Transport of Heat and Mass with Electrical Field - from Earth to Space." 2019 IEEE 20th International Conference on Dielectric Liquids (ICDL), Rome, Italy, June 23-27, 2019. 2019 IEEE 20th International Conference on Dielectric Liquids (ICDL), Rome, Italy, June 23-27, 2019. 5 p. https://doi.org/10.1109/ICDL.2019.8796729 , Jun-2019